Siying Chen

EDUCATION

Duke University Sep. 2024 – May 2026
MSc, Electrical and Computer Engineering Durham, NC
- Cumulative GPA: 3.63/4.0
- Relevant Coursework: Full-Stack IoT Systems, Performance Optimization & Parallelism, Random Signals and Noise

Macau University of Science and Technology Sep. 2020 – Aug. 2024
BSc, Computer Science Macau S.A.R., China
- Cumulative GPA: 3.87/4.0 (**Rank: 1/136**)
- Relevant Coursework: C, C++, Design and Analysis of Algorithms, Data Structures, Data Science, Operating System

RESEARCH INTERESTS

Contact-rich Manipulation, Sim-to-Real Transfer, Task-relevant Representation Learning, Multimodal Perception

RESEARCH EXPERIENCE

Task-Relevant Contact Representation for Contact-Rich Robotic Assembly Aug. 2026 – Present
Research Project in DexLab, advised by Prof. Xianyi Cheng Durham, NC
- Investigating which aspects of connector-level contact fidelity are necessary for robust robotic assembly and sim-to-real policy transfer, using LEGO Technic connectors as controlled contact-rich tasks.
- Developing task-relevant contact representations that abstract dense point contacts into compact geometric and physical interaction constraints, with an emphasis on generalization across connector geometries.
- Designing sim-to-real experiments to evaluate how contact representations and force–displacement fidelity affect force-aware manipulation policies and downstream assembly performance.

Can Vision-Language Models Safely Rate Robotic Surgical Skill? A Failure Mode Analysis with Video, Kinematics, and Expert References May 2026 – Aug. 2026
Research Project in DexLab, advised by Prof. Xianyi Cheng Durham, NC
- Investigated whether general-purpose VLMs can assess robotic surgical skill from JIGSAWS surgical videos with expert-rated modified OSATS and GRS scores.
- Built a controlled evaluation pipeline comparing vision only, kinematics only, raw kinematic grounding, interpreted grounding, and prompt variants across GPT, Claude, and Gemini.
- Analyzed metrics including Spearman correlation, MAE, and ICC, finding that GPT vision only at 3 fps performed best while grounding did not reliably improve expert agreement.

Evaluating Self-Supervised Pretraining for Wrist-Worn Physiological Signals in Cardiac Estimation and Gesture Recognition Nov. 2025 – Apr. 2026
Co-First Author, CHASE 2026 Durham, NC
- Designed a 4-channel EXG wearable platform with ESP32-S3 and EXG pills for simultaneous hand-gesture recognition and cardiac monitoring across **20 subjects, 9 gestures**.
- Built an ECG-analysis pipeline achieving **95.5% heart-rate accuracy within 5 bpm** of chest-lead ground truth and **89% HRV agreement** during active hand movements.
- Built a deep-learning DSP pipeline with STFT/CWT/LogMel transforms and ResNet18, achieving **79.1% gesture-classification accuracy** on forearm EMG channels.
- Implemented ESP-IDF firmware with FreeRTOS task distribution for real-time multi-sensor data acquisition.
- Designed a stimulus experiment with a custom PyQt data-acquisition UI for real-time multi-sensor recording.

PUBLICATIONS

[1] **S. Chen**, K. Geng, R. Liu, T. Chen, and J. Nie, "Evaluating Self-Supervised Pretraining for Wrist-Worn Physiological Signals in Cardiac Estimation and Gesture Recognition," accepted at *IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE 2026)*, 2026.

[2] **S. Chen** and X. Cheng, "Can Vision-Language Models Safely Rate Robotic Surgical Skill? A Failure Mode Analysis with Video, Kinematics, and Expert References," accepted at *MICCAI 2026 Workshop on Safe Surgery (SafeSurg)*, 2026. **(Second Best Paper Award)**

[3] Y. Hu, Z. Jia, **S. Chen**, D. Kelly, M. Zhao, T. Chen, R. Liu, and J. Nie, "Comparing Self-Supervised Pretraining Methods and Transfer Configurations for Wrist ECG under Motion," *The 4th International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys 2026)*, 2026.

WORK EXPERIENCE

Reality Inc. Jun. 2026 – Present
Embedded System Engineer Durham, NC

- Integrated an iPhone and Raspberry Pi-based AI call-agent prototype using Bluetooth HFP/SCO for cellular call control and audio, and BLE L2CAP CoC for bidirectional 8 kHz PCM transport.
- Developed and debugged Swift and Python components across the iOS app and Linux box, including BLE transport, Gemini Live audio streaming, automatic call answering, reconnect handling, and persistent systemd services.
- Designed a reproducible workflow to evaluate audio packetization, latency, queue behavior, and Bluetooth coexistence during live cellular calls, using end-to-end instrumentation to distinguish software limitations from hardware constraints.
- Established a remote build and deployment workflow using Xcode command-line tools, TestFlight, and Linux system services to support rapid iteration and hardware-in-the-loop testing.

Excelland Technology Co., Ltd. Jun. 2023 – Aug. 2023
Embedded Software Engineer Shenzhen, China

- Developed low-level C firmware for an STM32-based communications dongle, bridging a delivery robot and its PC controller over BLE and UART to enable fully wireless control.
- Designed and implemented a custom CAN-bus protocol for structured, reliable multi-node communication, defining message framing and node addressing.
- Integrated the dongle as a heartbeat node within the CAN bus to monitor node activity and ensure system reliability.
- Implemented a transport-layer filtering mechanism on the dongle, including serial commands to configure filtering parameters and operational modes.

AWARDS & HONORS

- **Second Best Paper Award**, MICCAI 2026 Workshop on Safe Surgery (SafeSurg), 2026
- Duke University Graduate Admission Scholarship, 2024
- Dean's Honor List, 2020–2024
- IET 2022 Student Prize, 2022
- ICBC Macau Scholarship, 2020/2021

SKILLS

Programming: Python, C/C++, MATLAB
Robotics & ML: PyTorch, OpenCV, MuJoCo, Newton
Embedded & Systems: Linux, STM32, ESP32, FreeRTOS, BLE, CAN, UART, Git